Determinants of Wealth and Financial Distress: A Panel Data and Binary Response Approach

1. Introduction

This report explores the determinants of household wealth and financial distress using both panel data and binary response models. Based on a longitudinal dataset spanning nine survey waves, the analysis examines how psychological distress and income influence economic outcomes, while controlling for demographic, employment, and life event factors. A combination of OLS, fixed effects, random effects, and logistic regression models is employed to capture both continuous variation in net worth and binary financial hardship. This dual-model approach offers a structured framework for assessing both the magnitude and probability of financial vulnerability across population subgroups.

2. Data Description and Summary Statistics

2.1 Data Source and Structure

The dataset consists of 210,390 observations and 27 variables. The panel structure is defined by two key identifiers: id (individual) and year (wave number, ranging from 0 to 8).

Key variables include:

> ***Dependent variable***: wealth – net worth (assets minus liabilities)
>
> ***Key independent variable***: `pd` – psychological distress (score from 0 to 24)
>
> ***Control variables***: age, gender, income, education, employment, life events, and ethnicity

Prior to analysis, three transformations were applied:

SAS code:

```
data work.data_transformed;
   set work.assessment2_data_withhead;
        ARSINH_wealth = ARSINH(wealth);
   age2 = age * age;
   logincome = log(income);
run;
```

ARSINH_wealth was computed using inverse hyperbolic sine to normalize extreme wealth values.

logincome was derived by log-transforming income to reduce skewness.

age2 was created as a squared term to capture potential non-linear age effects.

To ensure model validity, observations were restricted to household heads (head = 1) who had not changed households (notmoved = 1) or family composition (nofamichange = 1). Additionally, rows with missing values in any model-relevant variable were removed. This resulted in a cleaned panel dataset (panel_ready) used in subsequent regressions.

SAS code:

```
data work.panel_ready;
   set work.data_transformed;
   if head=1 and notmoved=1 and nofamichange=1
     and wealth~=. and pd~=. and age~=. and education~=. and income~=.
     and employed~=. and male~=. and divorce~=. and marriage~=.
     and childbirth~=. and familydeath~=. and laidoff~=. and missedwork~=.
     and white~=. and black~=. and hispanic~=. and logincome~=. and age2~=. ;
```

```
run;
```

2.2 Summary Statistics

SAS code:

```
proc means data=work.panel_ready n mean std min max;
   var wealth pd logincome age;
run;
proc freq data=work.panel_ready;
   tables male divorce employed white black hispanic;
run;
```

Summary table of means and standard deviations

The MEANS Procedure

Variable	Label	N	Mean	Std Dev	Minimum	Maximum
wealth	wealth	60413	263664.70	1233201.50	-3197000.00	100555000
pd	pd	60413	3.3318657	3.9553336	0	24.0000000
logincome		60413	2.3615794	0.1096381	0	2.7725887
age	age	60413	45.5981494	15.8875343	16.0000000	99.0000000

Frequency tables for binary variables

male				
male	Frequency	Percent	Cumulative Frequency	Cumulative Percent
0	18218	30.16	18218	30.16
1	42195	69.84	60413	100.00

divorce				
divorce	Frequency	Percent	Cumulative Frequency	Cumulative Percent
0	57549	95.26	57549	95.26
1	2864	4.74	60413	100.00

employed				
employed	Frequency	Percent	Cumulative Frequency	Cumulative Percent
0	16726	27.69	16726	27.69
1	43687	72.31	60413	100.00

white				
white	Frequency	Percent	Cumulative Frequency	Cumulative Percent
0	24858	41.15	24858	41.15
1	35555	58.85	60413	100.00

black				
black	Frequency	Percent	Cumulative Frequency	Cumulative Percent
0	38985	64.53	38985	64.53
1	21428	35.47	60413	100.00

hispanic				
hispanic	Frequency	Percent	Cumulative Frequency	Cumulative Percent
0	59820	99.02	59820	99.02
1	593	0.98	60413	100.00

The descriptive statistics show:

Average net worth is $263,664.70 with a large standard deviation of over $1.2 million, suggesting high variability and potential outliers.

Psychological distress (pd) has a mean of 3.33 and spans the full 0–24 range.

The sample is mostly male (69.84%), employed (72.31%), and white (58.85%).

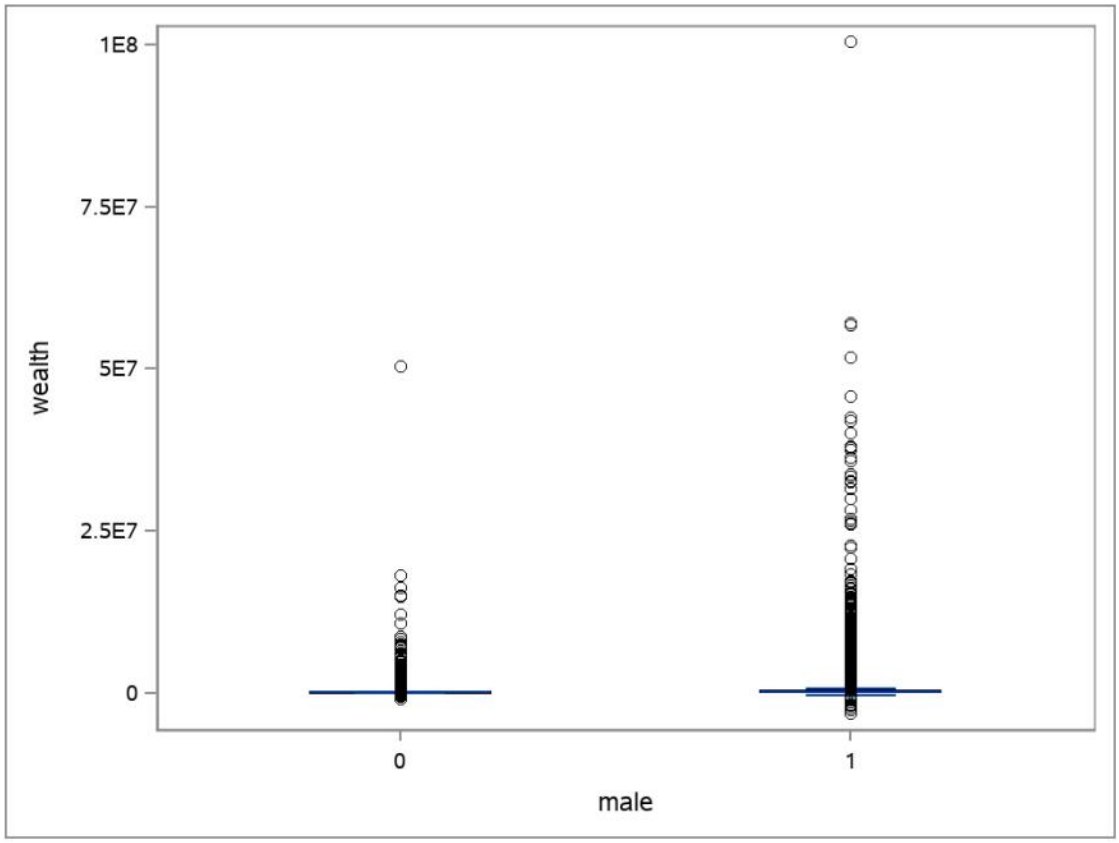
These patterns reflect structural characteristics of the dataset and help motivate the use of controls in subsequent models.

2.3 Initial Insights and Visualization

SAS code:

```
proc sgplot data=work.panel_ready;
    vbox wealth / category=male;
run;
```

Plot of wealth by gender



The plot of net worth by gender reveals that male respondents tend to have a wider range of wealth and more extreme outliers compared to females. While the median wealth appears similar across genders, the larger spread among males provides important context when interpreting regression results and suggests why gender is included as a control variable.

3. Panel Data Models on Net Worth

Model Specifications

3.1 Model 1a: OLS Regression

SAS code:

```
proc reg data=work.panel_ready;
   model ARSINH_wealth = pd male divorce marriage education logincome
                         age age2 employed childbirth familydeath laidoff missedwork
                         white black hispanic;
run;
quit;
```

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	16	824396	51525	963.56	<.0001
Error	60396	3229564	53.47314		
Corrected Total	60412	4053960			

Root MSE	7.31253	R-Square	0.2034
Dependent Mean	7.85928	Adj R-Sq	0.2031
Coeff Var	93.04331		

Parameter Estimates						
Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	Intercept	1	-32.51952	0.75167	-43.26	<.0001
pd	pd	1	-0.15313	0.00787	-19.44	<.0001
male	male	1	1.74514	0.07323	23.83	<.0001
divorce	divorce	1	-1.17837	0.14224	-8.28	<.0001
marriage	marriage	1	-0.13034	0.11802	-1.10	0.2695
education	education	1	-0.02171	0.01258	-1.73	0.0843
logincome		1	13.17988	0.34272	38.46	<.0001
age	age	1	0.21479	0.01108	19.38	<.0001
age2		1	-0.00070892	0.00011109	-6.38	<.0001
employed	employed	1	0.80523	0.08182	9.84	<.0001
childbirth	childbirth	1	0.21687	0.09632	2.25	0.0244
familydeath	familydeath	1	0.61146	0.21546	2.84	0.0045
laidoff	laidoff	1	-0.49401	0.13788	-3.58	0.0003
missedwork	missedwork	1	-0.05074	0.00824	-6.16	<.0001
white	white	1	0.90249	0.14333	6.30	<.0001
black	black	1	-0.93080	0.14777	-6.30	<.0001
hispanic	hispanic	1	1.44450	0.33234	4.35	<.0001

Interpretation – Model 1a: OLS Regression (PROC REG)

Model 1a shows that being **male** is associated with significantly higher transformed wealth (β = 1.7451, $p < .0001$), suggesting gender disparities in outcomes. The effect of **logincome** is also large and highly significant (β = 13.18), confirming that higher income strongly predicts higher wealth.

The coefficient for **pd** (psychological distress) is negative and significant (β = -0.1531, $p < .0001$), indicating a potential economic impact of mental health. **Hispanic** participants show significantly higher wealth (β = 1.4445), but the small sample size warrants caution.

Several variables (e.g., **marriage**, **education**) are not statistically significant. Others like **divorce** and **layoff** show negative and significant associations with wealth, consistent with life disruption effects.

The overall model is significant ($F(16, 60396) = 963.56$, $p < .0001$), with an **R^2 of 0.2034**, indicating that about 20% of the variation in transformed wealth is explained.

3.2 Model 1b: Pooled OLS with Panel Structure (PROC PANEL / POOLED)

SAS code:

```
proc panel data=work.panel_ready;
   id id year;
   model ARSINH_wealth = pd male divorce marriage education logincome
                         age age2 employed childbirth familydeath laidoff missedwork
                         white black hispanic / pooled;
run;
```

Model Description	
Estimation Method	Pooled
Number of Cross Sections	12455
Time Series Length	9

Fit Statistics			
SSE	3229563.963	DFE	60396
MSE	53.4731	Root MSE	7.3125
R-Square	0.2034		

Parameter Estimates

Variable	DF	Estimate	Standard Error	t Value	Pr > |t|	Label
Intercept	1	-32.5195	0.7517	-43.26	<.0001	Intercept
pd	1	-0.15313	0.00787	-19.44	<.0001	pd
male	1	1.745138	0.0732	23.83	<.0001	male
divorce	1	-1.17837	0.1422	-8.28	<.0001	divorce
marriage	1	-0.13034	0.1180	-1.10	0.2695	marriage
education	1	-0.02171	0.0126	-1.73	0.0843	education
logincome	1	13.17988	0.3427	38.46	<.0001	
age	1	0.214788	0.0111	19.38	<.0001	age
age2	1	-0.00071	0.000111	-6.38	<.0001	
employed	1	0.805233	0.0818	9.84	<.0001	employed
childbirth	1	0.216867	0.0963	2.25	0.0244	childbirth
familydeath	1	0.611463	0.2155	2.84	0.0045	familydeath
laidoff	1	-0.49401	0.1379	-3.58	0.0003	laidoff
missedwork	1	-0.05074	0.00824	-6.16	<.0001	missedwork
white	1	0.902487	0.1433	6.30	<.0001	white
black	1	-0.9308	0.1478	-6.30	<.0001	black
hispanic	1	1.444504	0.3323	4.35	<.0001	hispanic

Interpretation – Model 1b: Pooled OLS

Pooled OLS was applied to the panel data, treating it as a single cross-sectional dataset (*Cameron & Trivedi, 2005*). It yields coefficient estimates identical to standard OLS (Model 1a). For example, the effect of **pd** remains at **-0.15313**, with the same significance.

Since no individual or time effects are controlled, the model offers no new insights beyond Model 1a. However, it serves as a baseline for comparing with fixed and random effects models that do account for unobserved heterogeneity.

3.3 Model 2: Fixed Effects (Entity) — /FIXONE

SAS code:

```
proc panel data=work.panel_ready;
```

```
    id id year;
    class year male;
    model ARSINH_wealth = pd male divorce marriage education logincome
                          age age2 employed childbirth familydeath laidoff missedwork
                          white black hispanic
                          / fixone;
run;
```

Model Description	
Estimation Method	FixOne
Number of Cross Sections	12455
Time Series Length	9

Fit Statistics			
SSE	1463889.664	DFE	47943
MSE	30.5340	Root MSE	5.5258
R-Square	0.6389		

F Test for No Fixed Effects			
Num DF	Den DF	F Value	Pr > F
12454	47943	4.64	<.0001

Parameter Estimates						
Variable	DF	Estimate	Standard Error	t Value	Pr > |t|	Label
Intercept	1	-16.6214	2.9776	-5.58	<.0001	Intercept
pd	1	-0.03318	0.00924	-3.59	0.0003	pd
MALE1	0	0	.	.	.	MALE = 1
MALE0	0	0	.	.	.	MALE = 0
divorce	1	-0.71518	0.1355	-5.28	<.0001	divorce
marriage	1	0.280332	0.1132	2.48	0.0133	marriage
education	1	-0.07522	0.0395	-1.90	0.0569	education
logincome	1	2.751593	0.3834	7.18	<.0001	
age	1	0.209945	0.0203	10.34	<.0001	age
age2	1	-0.0016	0.000201	-7.94	<.0001	
employed	1	0.352471	0.0856	4.12	<.0001	employed
childbirth	1	0.080311	0.0967	0.83	0.4064	childbirth
familydeath	1	0.045289	0.2033	0.22	0.8237	familydeath
laidoff	1	0.05584	0.1394	0.40	0.6887	laidoff
missedwork	1	-0.02167	0.00804	-2.69	0.0071	missedwork
white	1	0.39148	0.3521	1.11	0.2663	white
black	1	0.061316	0.5655	0.11	0.9137	black
hispanic	1	1.228136	0.3622	3.39	0.0007	hispanic

Interpretation – Model 2: Fixed Effects (Entity)

Compared to the pooled model (Model 1b), the fixed effects model (Model 2) introduces individual-specific intercepts to control for unobserved heterogeneity

across individuals. This significantly improves explanatory power: **R-squared increases from 0.2034 to 0.6389**, indicating that individual fixed effects play a substantial role in explaining the variation in transformed wealth.

Notably, key coefficients change. The effect of **pd** (psychological distress) **shrinks** from -0.1531 to **-0.0332**, though it remains significant (p = 0.0003). This implies prior results were likely inflated by omitted individual effects.

Gender variables (MALE1 and MALE0) are not estimated in Model 2 because they are time-invariant. Their effects are absorbed into the individual fixed intercepts and thus not shown as separate coefficients. Fixed effects absorb all time-invariant heterogeneity, such as gender and ethnicity. (*Baltagi, 2008*)

The effect of **laidoff** becomes statistically insignificant, suggesting its earlier association may reflect between-person differences rather than a within-person causal effect. The impact of **logincome** also declines substantially (from 13.18 to 2.75). Life events like **divorce** and **missedwork** remain negatively associated with wealth, though with smaller effects, while **marriage** becomes significant in Model 2 (p = 0.0133), unlike in Model 1b.

The **F test for fixed effects** is strongly significant (F = 4.64, p < .0001), confirming the joint importance of fixed effects and the need to control for unobserved heterogeneity.

3.4 Model 3: Fixed Effects (Time) — /FIXTWO

SAS code:

```
proc panel data=work.panel_ready;
   id id year;
   class year male;
   model ARSINH_wealth = pd male divorce marriage education logincome
                age age2 employed childbirth familydeath laidoff missedwork
                white black hispanic
                /fixtwo;
run;
```

Model Description	
Estimation Method	FixTwo
Number of Cross Sections	12455
Time Series Length	9

Fit Statistics			
SSE	1452744.412	DFE	47935
MSE	30.3065	Root MSE	5.5051
R-Square	0.6416		

F Test for No Fixed Effects			
Num DF	Den DF	F Value	Pr > F
12462	47935	4.70	<.0001

Parameter Estimates						
Variable	DF	Estimate	Standard Error	t Value	Pr > |t|	Label
Intercept	1	-11.0628	5.2571	-2.10	0.0354	Intercept
pd	1	-0.03253	0.00921	-3.53	0.0004	pd
MALE1	0	0	.	.	.	MALE = 1
MALE0	0	0	.	.	.	MALE = 0
divorce	1	-0.65297	0.1351	-4.83	<.0001	divorce
marriage	1	0.254393	0.1128	2.25	0.0242	marriage
education	1	-0.07148	0.0394	-1.81	0.0700	education
logincome	1	2.779547	0.3825	7.27	<.0001	
age	1	0.110348	0.0825	1.34	0.1810	age
age2	1	-0.00144	0.000201	-7.18	<.0001	
employed	1	0.292761	0.0855	3.42	0.0006	employed
childbirth	1	0.104655	0.0964	1.09	0.2776	childbirth
familydeath	1	0.055458	0.2025	0.27	0.7842	familydeath
laidoff	1	0.102212	0.1391	0.73	0.4625	laidoff
missedwork	1	-0.02138	0.00801	-2.67	0.0076	missedwork
white	1	0.438032	0.3509	1.25	0.2119	white
black	1	0.083142	0.5635	0.15	0.8827	black
hispanic	1	0.868568	0.3632	2.39	0.0168	hispanic

Interpretation – Model 3: Fixed Time Effects (FixTwo)

This model incorporates time-specific intercepts. The **R-squared improves to 0.6416**, indicating that **time effects explain substantial variation in wealth**, above the pooled model (0.2034) and close to the entity fixed effects model (0.6389).

The coefficient for **pd** (–0.0325), nearly identical to Model 2 and far smaller than in the pooled model, reinforcing the presence of unobserved structural factors. The coefficient for **logincome** remains stable between the fixone and fixtwo models.

Marriage remains positive and significant (p = 0.0242), while **laid-off** and **black** continue to show no significance. **White** remains insignificant, and **hispanic** stays

significant but with a reduced coefficient. **Education** is marginally significant in Model 3 (p = 0.0700), weaker than in Model 2, suggesting a less robust association with wealth.

The **F test is significant (F = 4.70, p < .0001)**, validating the inclusion of time effects and supporting the robustness of key predictors like **pd.**

3.5 Model 4: Random Effects (Entity) — /RANONE

SAS code :

```
proc panel data=work.panel_ready;
   id id year;
   class year male;
   model ARSINH_wealth = pd male divorce marriage education logincome
                         age age2 employed childbirth familydeath laidoff missedwork
                         white black hispanic
                         / ranone;
run;
```

Fit Statistics			
SSE	1819438.501	DFE	60396
MSE	30.1251	Root MSE	5.4886
R-Square	0.0641		

Variance Component Estimates	
Variance Component for Cross Sections	30.5245
Variance Component for Error	30.53396

Hausman Test for Random Effects			
Coefficients	DF	m Value	Pr > m
15	15	827.90	<.0001

Parameter Estimates						
Variable	DF	Estimate	Standard Error	t Value	Pr > |t|	Label
Intercept	1	-13.5392	0.8252	-16.41	<.0001	Intercept
pd	1	-0.07924	0.00817	-9.70	<.0001	pd
MALE1	0	0	.	.	.	MALE = 1
MALE0	1	-2.12806	0.1257	-16.93	<.0001	MALE = 0
divorce	1	-0.79031	0.1247	-6.34	<.0001	divorce
marriage	1	0.167789	0.1044	1.61	0.1081	marriage
education	1	-0.00607	0.0199	-0.30	0.7605	education
logincome	1	5.88323	0.3407	17.27	<.0001	
age	1	0.229306	0.0143	16.02	<.0001	age
age2	1	-0.00108	0.000144	-7.52	<.0001	
employed	1	0.619101	0.0785	7.89	<.0001	employed
childbirth	1	0.181214	0.0894	2.03	0.0426	childbirth
familydeath	1	0.362857	0.1881	1.93	0.0537	familydeath
laidoff	1	-0.08088	0.1287	-0.63	0.5298	laidoff
missedwork	1	-0.02857	0.00751	-3.81	0.0001	missedwork
white	1	0.837753	0.2107	3.98	<.0001	white
black	1	-1.43787	0.2220	-6.48	<.0001	black
hispanic	1	1.851348	0.3057	6.06	<.0001	hispanic

Interpretation – Model 4: Random Effects (Entity)

This model estimates random intercepts for individuals, allowing time-invariant predictors such as gender to be included. $\mathbf{R^2}$ is low (0.0641), but key effects remain significant. **Psychological distress (pd)** negatively predicts wealth (β = –0.0792), and **logincome** remains a strong positive driver (β = 5.88; both p < .0001).

Gender disparities re-emerge: **Females (MALE = 0)** show significantly lower wealth (β = –2.13, p < .0001), a gap not estimable in fixed effects models. Unlike fixed effects models, random effects allow estimation of time-invariant predictors like **white** and **black**, revealing significant racial disparities in wealth. Specifically, white participants show significantly higher wealth than Black participants (β = 0.84 vs –1.44; both p < .0001),

Model 4 confirms the importance of income, gender, and mental health, while showing the flexibility of random effects in capturing individual heterogeneity.

3.6 Model 5: Two-Way Random Effects — /RANTWO

SAS code :

```
proc panel data=work.panel_ready;
   id id year;
   class year male;
   model ARSINH_wealth = pd male divorce marriage education logincome
                         age age2 employed childbirth familydeath laidoff missedwork
                         white black hispanic
                         / rantwo;
run;
```

Fit Statistics			
SSE	1763243.760	DFE	60396
MSE	29.1947	Root MSE	5.4032
R-Square	0.0627		

Variance Component Estimates	
Variance Component for Cross Sections	35.08777
Variance Component for Time Series	0.560741
Variance Component for Error	30.30655

Hausman Test for Random Effects			
Coefficients	DF	m Value	Pr > m
15	15	469.40	<.0001

Parameter Estimates						
Variable	DF	Estimate	Standard Error	t Value	Pr > |t|	Label
Intercept	1	-15.2059	0.8751	-17.38	<.0001	Intercept
pd	1	-0.0741	0.00813	-9.11	<.0001	pd
MALE1	0	0	.	.	.	MALE = 1
MALE0	1	-2.14994	0.1308	-16.44	<.0001	MALE = 0
divorce	1	-0.76598	0.1237	-6.19	<.0001	divorce
marriage	1	0.134066	0.1035	1.30	0.1952	marriage
education	1	0.046826	0.0207	2.26	0.0236	education
logincome	1	6.003185	0.3398	17.67	<.0001	
age	1	0.247337	0.0145	17.00	<.0001	age
age2	1	-0.00105	0.000146	-7.22	<.0001	
employed	1	0.486186	0.0781	6.23	<.0001	employed
childbirth	1	0.144348	0.0887	1.63	0.1037	childbirth
familydeath	1	0.226869	0.1864	1.22	0.2237	familydeath
laidoff	1	-0.1401	0.1280	-1.09	0.2738	laidoff
missedwork	1	-0.02933	0.00744	-3.94	<.0001	missedwork
white	1	0.720292	0.2147	3.36	0.0008	white
black	1	-1.33394	0.2269	-5.88	<.0001	black
hispanic	1	0.911651	0.3083	2.96	0.0031	hispanic

Interpretation – Model 5: Two-Way Random Effects (Entity + Time)

This model adds random intercepts for individuals and time. The $\mathbf{R^2}$ is 0. 0627, slightly below the one-way random effects model (Model 4: 0.0641), and substantially below fixed effects models (Model 2 and 3, both above 0.63). The additional time component contributes a small variance (0.56), indicating limited year-to-year fluctuation in wealth once individual variation is considered.

pd remains a strong and negative predictor of wealth (β = –0.0741, p < .0001), consistent across all models. **logincome** shows a strong effect here (β = 6.00, p

< .0001). **MALE = 0** remains strongly negative (β = –2.15), reaffirming a gender wealth gap.

Notably, **education** becomes significant (p = 0.0236) in this model, which is a departure from Models 1–4 where it was either insignificant or marginal. This may suggest that when both entity- and time-level random effects are accounted for, the role of education becomes more discernible, possibly because the interaction between time and individual-level unobserved heterogeneity enhances its statistical power.

Other effects, such as **marriage**, **familydeath**, **laidoff**, and **childbirth**, are statistically insignificant, though their directionality remains consistent with economic expectations.

Ethnicity continues to play a role: **Black** individuals have significantly lower wealth (β = –1.33, p < .0001), while **Hispanic** individuals have higher wealth (β = 0.91, p = 0.0031), though the magnitude is reduced compared to Model 4.

3.7 Model Selection: Hausman Test

SAS code :

```
proc panel data=work.panel_ready outest=fixed_out;
   id id year;
   class year male;
   model ARSINH_wealth = pd male divorce marriage education logincome
                         age age2 employed childbirth familydeath laidoff missedwork
                         white black hispanic
                         / fixone;
run;
proc panel data=work.panel_ready outest=random_out;
   id id year;
   class year male;
```

```
    model ARSINH_wealth = pd male divorce marriage education logincome
                          age age2 employed childbirth familydeath laidoff missedwork
                          white black hispanic
                          / ranone;
run;
```

Hausman Test for Random Effects			
Coefficients	DF	m Value	Pr > m
15	15	827.90	<.0001

To determine whether fixed or random effects are more appropriate, a Hausman test was conducted (*Wooldridge, 2010; Greene, 2012*) comparing Model 2 (/fixone) and Model 4 (/ranone). The test returned a highly significant result ($\chi^2 = 827.90$, df = 15, $p < .0001$), rejecting the null hypothesis that the random effects estimator is consistent. Consequently, the fixed effects specification is preferred for consistent estimation of coefficients. The result supports the use of Model 2 or Model 3 over random effects models for analysing net wealth outcomes in this dataset.

4. Binary Response Model: Financial Distress

SAS code :

```
proc logistic data=work.panel_ready descending;
    class year male;
    model financial_distress = pd male divorce marriage education logincome
                               age age2 employed childbirth familydeath laidoff missedwork
                               white black hispanic;
run;
```

Analysis of Maximum Likelihood Estimates						
Parameter		DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept		1	1.8350	0.2776	43.6864	<.0001
pd		1	0.0451	0.00289	244.5184	<.0001
male	0	1	0.2338	0.0140	280.6052	<.0001
divorce		1	0.1425	0.0493	8.3546	0.0038
marriage		1	0.1067	0.0420	6.4373	0.0112
education		1	0.1017	0.00575	313.1450	<.0001
logincome		1	-1.5759	0.1230	164.2735	<.0001
age		1	-0.0165	0.00533	9.6058	0.0019
age2		1	-0.00029	0.000060	23.6817	<.0001
employed		1	-0.0877	0.0326	7.2509	0.0071
childbirth		1	0.0273	0.0359	0.5803	0.4462
familydeath		1	-0.1536	0.1059	2.1051	0.1468
laidoff		1	0.1999	0.0510	15.3738	<.0001
missedwork		1	0.0209	0.00290	51.9387	<.0001
white		1	-0.1788	0.0581	9.4747	0.0021
black		1	0.0408	0.0589	0.4794	0.4887
hispanic		1	-0.6126	0.1682	13.2661	0.0003

Odds Ratio Estimates			
Effect	Point Estimate	95% Wald Confidence Limits	
pd	1.046	1.040	1.052
male 0 vs 1	1.596	1.511	1.686
divorce	1.153	1.047	1.270
marriage	1.113	1.025	1.208
education	1.107	1.095	1.120
logincome	0.207	0.163	0.263
age	0.984	0.973	0.994
age2	1.000	1.000	1.000
employed	0.916	0.859	0.976
childbirth	1.028	0.958	1.103
familydeath	0.858	0.697	1.055
laidoff	1.221	1.105	1.350
missedwork	1.021	1.015	1.027
white	0.836	0.746	0.937
black	1.042	0.928	1.169
hispanic	0.542	0.390	0.754

A logistic regression showed that several predictors have clear directional effects on financial distress. For example, **pd** increases the odds of financial hardship (**OR = 1.046**, $p < .0001$), or about 4.6% per unit increase. In contrast, higher **logincome** substantially reduces the odds by approximately 79.3% (**OR = 0.207**, $p < .0001$).

Male respondents also face significantly higher risk: being male increases the odds of hardship by about 59.6% compared to females (OR = 1.596, $p < .0001$). **Hispanic** respondents are less likely to report financial distress.

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	4792.9947	16	<.0001
Score	4647.0749	16	<.0001
Wald	3948.7731	16	<.0001

The model fit is strong ($\chi^2 = 4792.99$, $p < .0001$), and key effects align with patterns observed in panel wealth models.

Association of Predicted Probabilities and Observed Responses			
Percent Concordant	72.6	Somers' D	0.452
Percent Discordant	27.4	Gamma	0.452
Percent Tied	0.0	Tau-a	0.112
Pairs	453466290	c	0.726

The model demonstrates acceptable discrimination, with a **c-statistic** of 0.726, meaning it correctly ranks distressed versus non-distressed cases in approximately 72.6% of paired comparisons.

5. Conclusion

Higher **psychological distress (pd)** is significantly associated with lower net wealth and higher odds of financial hardship. Across all panel models, higher pd levels are associated with reduced wealth, reinforcing the economic consequences of mental health issues.

The effect of income (**logincome**) remains statistically significant in all net worth models but varies in magnitude. Its predictive power diminishes in fixed effects models, suggesting that part of its influence may stem from unobserved individual or temporal heterogeneity.

Gender disparities are evident: female respondents consistently report lower wealth, while male respondents exhibit higher odds of financial hardship, reinforcing gender disparities observed in both wealth and distress outcomes. These patterns highlight the structural nature of gender-based financial inequality.

The **Hausman test** supports the use of fixed effects, rejecting the random effects assumption and confirming that individual-specific traits are correlated with regressors—warranting controls for consistent estimation.

Complementing the panel analysis, the logistic regression model shifts focus from wealth levels to the likelihood of hardship (*Long & Freese, 2014*). While overlapping predictors emerge—such as pd and employment—the logistic framework emphasizes their role in shaping financial risk thresholds rather than absolute outcomes.

By combining **continuous and binary** outcome models, the analysis provides a more nuanced perspective on how mental health and income relate to financial vulnerability in panel data.

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Generative AI use statement

I, Yalan Zhuang, hereby confirm that I used https://chatgpt.com to assist in the development of this data analysis report. The tool was used to clarify analytical concepts, improve the structure and readability of SAS code, and refine the wording of figure interpretations and final conclusions. All AI-assisted content was carefully reviewed and revised by me to ensure alignment with the project requirements and academic integrity.

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I used ChatGPT for error correction, concept clarification, and idea expansion. It also assisted me in critically refining my code for better integration and coherence. Additionally, I used the tool to help translate and distill text, as well as to better understand and apply information from relevant websites. The assignment was independently completed by me, with ChatGPT serving only as a supplementary tool. All judgments, analyses, and decisions were made solely by myself.